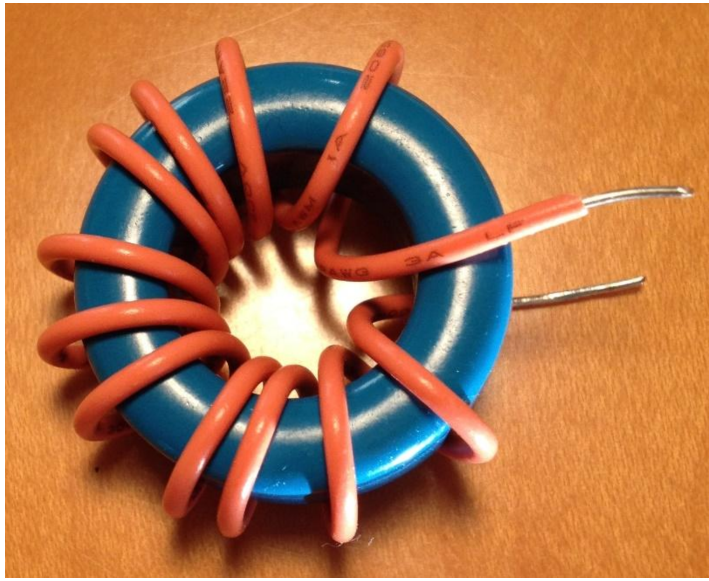
Theory behind winding inductors

- If a material is used to contain the magnetic flux:
  - Magnetic Field Strength = $\frac{NI}{\ell_e}$ (Amp Turns/meter)
    - N = number of turns, I = current, ℓ_e = effective length
  - Magnetic Flux Density = B = $\mu_r \mu_o \frac{NI}{\ell_e}$ (Teslas)
    - μ_r = relative permeability, μ_o= permeability of free space
  - Magnetic Flux = $N \mu_r \mu_o \frac{A_e}{\ell_e} I$ (Webers)
    - A_e = effective cross sectional area
  - Total Flux linked by coil = $\Phi = \overbrace{N^2 \underbrace{\mu_r \mu_o \frac{A_e}{\ell_e}}_{A_L}}^{L} I$
  - $L = N^2 A_L$ Core manufacturer specifies A_L (Henries/Turn^2)



Properties of magnetic materials

- Magnetic materials saturate. In most cases you don't want to exceed $B_{sat}/2$. This limits the maximum current. $B = \mu_r \mu_o \frac{NI}{\ell_e}$ (Teslas)
- $\mu_o = 1.26E - 6 \ \frac{Tesla \cdot meter}{Amp}$

Material	μ_r	$\boldsymbol{B_{sat}}$
Powdered Iron	~20	1T
Ferrite	~6000	0.4T
Steel	~5000	2T

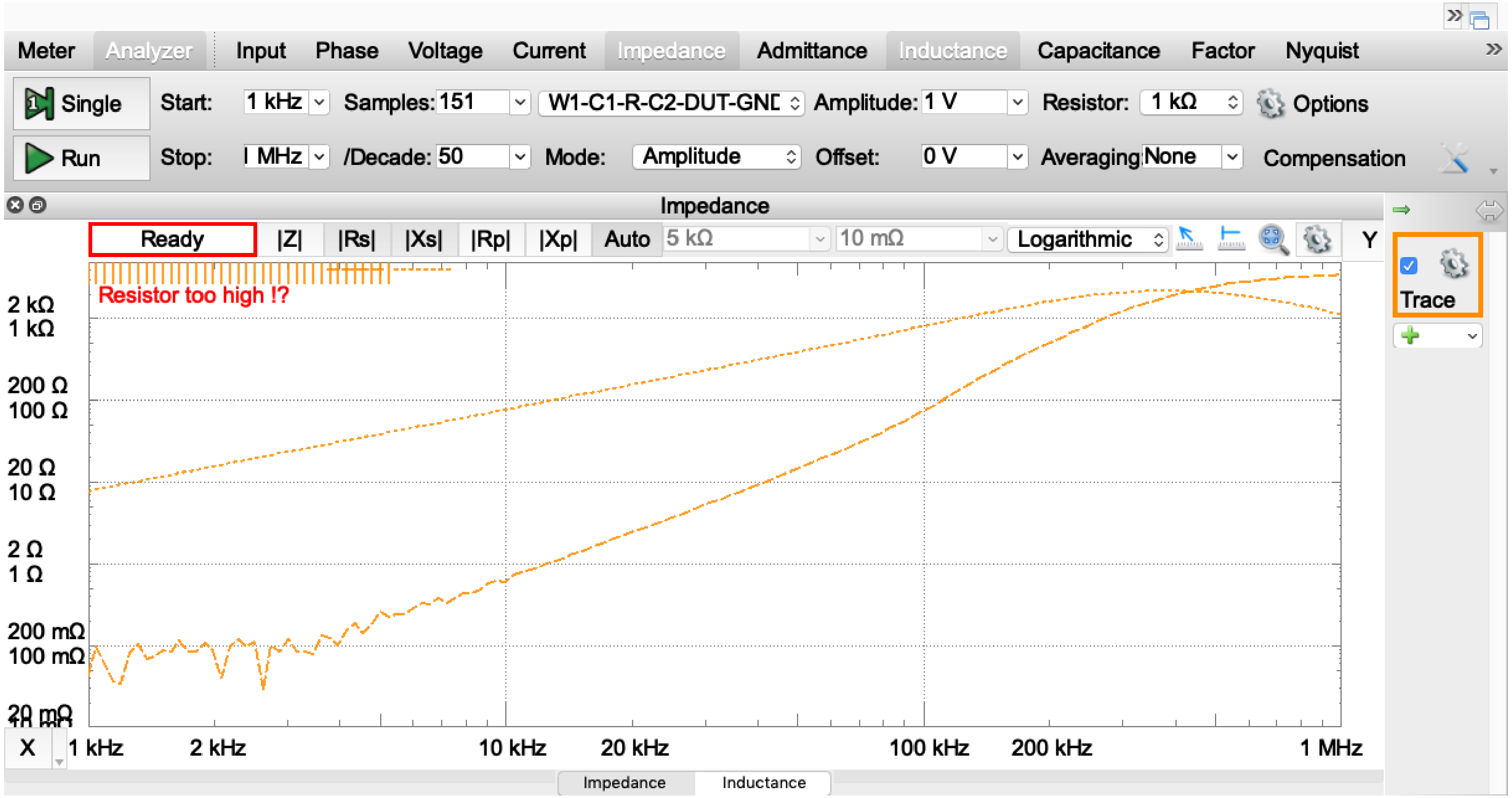
Inductor example: Make a 2mH inductor

- For our ferrite cores, the manufacturer claims $A_L = $ 6.9uH/turn^2
- $L = N^2 A_L$
- Count a turn for every time the wire goes through the center of the toroid. Can’t have fractional turns
- To figure out how much wire you need: 1.4”/turn + 3”

The AD2 enables us to automate the impedance measurement

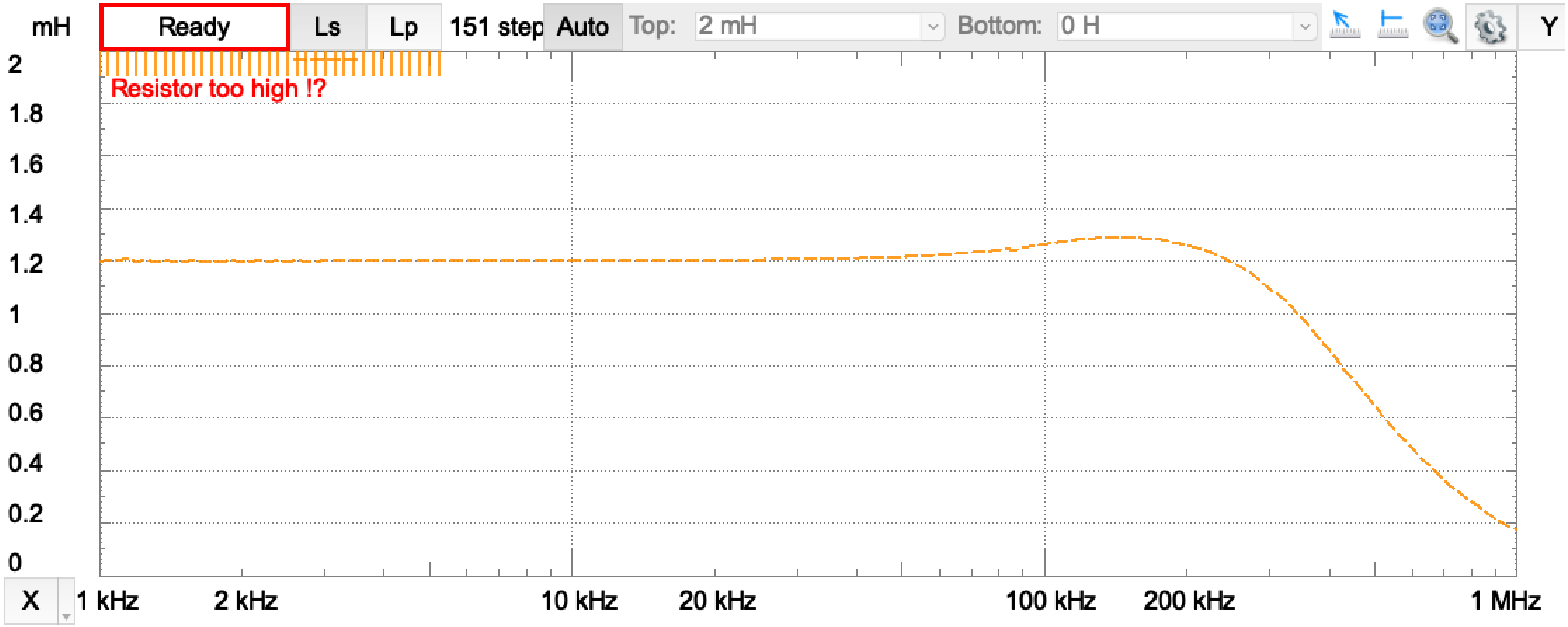
- Three configurations to choose from:
    - W1-C1-DUT-C2-R-GND
    - W1-C1-R-C2-DUT-GND
    - W1-C1P-DUT-C1N-C2-R-GND
- Must use a power of 10 resistance and specify it in the interface
- Can control the frequency range and amplitude of the input signal
- Many plotting options
    - Real and imaginary parts of impedance
    - Inductance or capacitance
    - Meter
- Watch time waveforms to make sure you aren’t getting junk

Inductor example: Impedance Measurement



- Make sure to select the resistor you use in your circuit
- Note that the imaginary component increases linearly (as expected)
- The resistive component also increases with frequency, most of which is loss, not winding resistance

Inductor example: Inductance Measurement



- When plotting series inductance, the inductance should be nearly constant below 100kHz
- The nearly constant value should match our computation
- The inductance will start to fall of after about 100kHz

Inductor example: Meter Measurement